

POC REQUIREMENTS DOCUMENT

Project Name: **Capacity & Utilization Intelligence Agent (CUIA)**

Project Type: Proof of Concept

Owner: Noel Mathews, Priyatham Sai Kethavarapu

Objective:

This Proof of Concept validates the ability to securely transform engineering execution data from Jira into actionable workforce capacity and utilization insights. It exists to prove that deterministic analytical models, combined with a secure, read-only AI Copilot, can effectively support engineering leadership in workload balancing and capacity forecasting without risking AI hallucinations or data exposure.

Tech Stack:

- Frontend: React
- Backend: FastAPI, PostgreSQL, Background Worker
- AI & Identity: LangGraph, Microsoft Entra ID
- Analytics: Deterministic Analytics (Python/Pandas)

In Scope:

- Entra ID Auth & Role-Based Access Control (RBAC).
- Jira data sync & CSV uploads (leave/skills).
- Immutable analytics snapshots (utilization, capacity, productivity).
- Role-scoped dashboards & rule-based recommendations.
- AI Copilot restricted to explaining pre-calculated data.
- Admin interface (configuration & data quality resolution).
- Audit logging for sensitive interactions.

Out of Scope:

- Multi-tenancy, high availability (HA), or CI/CD pipelines.
- Notifications (Email, Slack) or integrations beyond Jira.
- Mobile apps, autonomous AI agents, or AI-driven calculations/modifications.

FUNCTIONAL REQUIREMENTS:

- FR-1 Authenticate users exclusively via Entra ID.
- FR-2 Enforce strict backend RBAC for data visibility.
- FR-3 Ingest Jira data and allow CSV uploads for leave/skills.
- FR-4 Isolate malformed data without halting analytics runs.
- FR-5 Calculate and persist metrics deterministically as immutable snapshots.
- FR-6 Render role-aware executive and team dashboards.
- FR-7 Generate deterministic workforce recommendations.
- FR-8 Provide an AI Copilot that queries only snapshot data.
- FR-9 Provide an Admin interface for configurations and data resolution.
- FR-10 Maintain immutable audit logs for configurations and AI interactions.

NON-FUNCTIONAL REQUIREMENTS:

- Security & Auth: Zero-trust backend, default-deny RBAC, with authentication fully delegated to Entra ID.
- AI Boundaries: AI acts purely as an explanation layer, forbidden from calculating metrics or bypassing filters.
- Performance & Reliability: Async background processing with graceful degradation; dashboards consume fast, pre-calculated snapshots.
- Architecture & Deployment: Modular monolith deployed via a single Docker Compose command for rapid demonstration.

SUCCESS CRITERIA / DEFINITION OF DONE

- Users authenticate via Entra ID with correctly enforced roles.
- Jira data syncs and maps to internal users successfully.
- Background worker correctly generates immutable analytics snapshots.
- Dashboards display role-accurate utilization and capacity metrics.
- Admins can view and resolve flagged data quality issues.
- AI Copilot answers queries securely using only snapshot data.
- Admin workflows (uploads, config, manual triggers) function flawlessly.
- End-to-end demonstration runs sequentially without developer intervention.